

Engineering Mathematics Study Note: Directional Derivative Calculation

1. IDENTIFY

The concept being executed is the calculation of the **Directional Derivative** of a scalar field ϕ along an explicitly given vector direction $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ at a specific point in 3D space.

2. THE FORMULA

The Directional Derivative (D.D) of a scalar function ϕ along a vector direction $\mathbf{a}$ is given by:

$$\text{D.D} = \nabla\phi \cdot \hat{\mathbf{a}} = \frac{\nabla\phi \cdot \mathbf{a}}{|\mathbf{a}|}$$

Where the Gradient vector $\nabla\phi$ is:

$$\nabla\phi = \mathbf{i}\frac{\partial\phi}{\partial x} + \mathbf{j}\frac{\partial\phi}{\partial y} + \mathbf{k}\frac{\partial\phi}{\partial z}$$

And the unit direction vector $\hat{\mathbf{a}}$ is:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

3. THE ALGORITHM

Step 1: Compute the partial derivatives

Find the derivative of ϕ with respect to each variable (x , y , and z) one at a time while holding the remaining two variables constant.

$$\frac{\partial\phi}{\partial x}, \quad \frac{\partial\phi}{\partial y}, \quad \frac{\partial\phi}{\partial z}$$

Step 2: Construct the Gradient expression

Assemble the partial derivatives into a single three-dimensional vector using the directional unit vectors.

$$\nabla\phi = \mathbf{i}\frac{\partial\phi}{\partial x} + \mathbf{j}\frac{\partial\phi}{\partial y} + \mathbf{k}\frac{\partial\phi}{\partial z}$$

Step 3: Evaluate the Gradient at the given point

Substitute the numerical values of the coordinates $(x, y, z) = (2, -1, 2)$ into the variables of the gradient vector to get a fixed numerical vector.

Step 4: Calculate the magnitude of the direction vector

Find the total length of the provided directional vector $\mathbf{a}$ using the 3D Pythagorean distance formula.

$$|\mathbf{a}| = \sqrt{(a_x)^2 + (a_y)^2 + (a_z)^2}$$

Step 5: Perform the Dot Product and scale

Compute the dot product between the numerical gradient vector (from Step 3) and the directional vector $\mathbf{a}$. Divide that scalar result by the vector's magnitude (from Step 4).

4. THE MECHANICS

Resolving Algebraic Multipliers during Partial Differentiation

When a term contains multiple variables multiplied together, isolate the active variable and treat everything else as a single cohesive constant multiplier.

- **Example:** Differentiating $-3x^2y^2z$ with respect to x :
Treat the block $(-3y^2z)$ as a single constant number. Apply the power rule only to x^2 , which becomes $2x$. Then multiply them together:

$$\frac{\partial}{\partial x}(-3x^2y^2z) = (-3y^2z) \cdot (2x) = -6xy^2z$$

Evaluating Arithmetic with Negative Numbers

When substituting negative coordinate numbers like $y = -1$, keep them strictly enclosed in brackets to avoid sign placement errors, especially when dealing with powers.

- $(-1)^2 = 1$ (A negative number raised to an even power becomes positive)
- $-3(2)^2(2) \cdot 2(-1) = -3(4)(2) \cdot (-2) = -24 \cdot (-2) = +48$

5. STEP-BY-STEP WORKED SOLUTIONS

Execution of Step 1 & 2: Finding $\nabla\phi$ analytically

$$\begin{aligned}\frac{\partial\phi}{\partial x} &= \frac{\partial}{\partial x}(4xz^3 - 3x^2y^2z) = 4z^3 - 6xy^2z \\ \frac{\partial\phi}{\partial y} &= \frac{\partial}{\partial y}(4xz^3 - 3x^2y^2z) = 0 - 6x^2yz = -6x^2yz \\ \frac{\partial\phi}{\partial z} &= \frac{\partial}{\partial z}(4xz^3 - 3x^2y^2z) = 12xz^2 - 3x^2y^2\end{aligned}$$

Now, piece them together into the vector format:

$$\nabla\phi = \mathbf{i}(4z^3 - 6xy^2z) + \mathbf{j}(-6x^2yz) + \mathbf{k}(12xz^2 - 3x^2y^2)$$

Execution of Step 3: Evaluating $\nabla\phi$ at $(2, -1, 2)$

Substitute $x = 2$, $y = -1$, and $z = 2$:

$$\begin{aligned}\text{i-component} &= 4(2)^3 - 6(2)(-1)^2(2) = 4(8) - 6(2)(1)(2) = 32 - 24 = 8 \\ \text{j-component} &= -6(2)^2(-1)(2) = -6(4)(-1)(2) = 48 \\ \text{k-component} &= 12(2)(2)^2 - 3(2)^2(-1)^2 = 12(2)(4) - 3(4)(1) = 96 - 12 = 84\end{aligned}$$

Combine the evaluated components:

$$\nabla\phi_{(2,-1,2)} = 8\mathbf{i} + 48\mathbf{j} + 84\mathbf{k}$$

Execution of Step 4: Finding Magnitude of $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$

$$|\mathbf{a}| = \sqrt{2^2 + 3^2 + 6^2}$$

$$|\mathbf{a}| = \sqrt{4 + 9 + 36}$$

$$|\mathbf{a}| = \sqrt{49} = 7$$

Execution of Step 5: Dot Product and Final Division

$$\text{D.D} = \frac{(8\mathbf{i} + 48\mathbf{j} + 84\mathbf{k}) \cdot (2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k})}{7}$$

$$\text{D.D} = \frac{(8 \times 2) + (48 \times 3) + (84 \times 6)}{7}$$

$$\text{D.D} = \frac{16 + 144 + 504}{7}$$

$$\text{D.D} = \frac{664}{7}$$